

Bishop's University

CS 566: Advance Topics in Artificial Intelligence – Final Project

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Group:FP A

Q1-

```
PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code> C:\Python27\python.exe dataclassifier.py -c perceptron
Doing classification
-----
data:          digits
classifier:      perceptron
using enhanced features?: False
training set size: 100
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
55 correct out of 100 (55.0%).
Testing...
48 correct out of 100 (48.0%).
PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code>
```

Q2.

```
PS C:\Users\Ahmad\Desktop\MS_CS\2-Summer-24\CS566-01 202405 - Artificial Int
elligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -w
Doing classification
-----
data:          digits
classifier:     perceptron
using enhanced features?: False
training set size: 100
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
55 correct out of 100 (55.0%).
Testing...
48 correct out of 100 (48.0%).
=== Features with high weight for label 0 ===

      #  ##
    #####
    #####
    #####
          #
      ###  #
#   ##   ##
##      ##
```

```
#####
#####
#   ##  #
#####  #
#####
#### ##  #
#####
# #####  #
#  #  #
##   ##
```

=== Features with high weight for label 2 ===

```
#####
### #  #
#####
#####  #
###   ## ####
#    #    #
    ##    ##
        #
```

```
          #    #
        ##    #
#   ####    #
##   ####    # #
###   ### #   ###
##    ###    #
##    ##    #  #
#####        #
          #  #
        ###  #
```

```
55 correct out of 100 (55.0%).
Testing...
48 correct out of 100 (48.0%).
=== Features with high weight for label 0 ===
```

```
      #  ##
    #####
    #####
    #####
      #
    ###  #
  #  ##  ##
  ##     ##
  ##     ##
      ##
### #     ##
## #     ##
###      #
##### #  #  #
###
####      ##
#####
##### #
# #####
#####
  ##
```

```
=== Features with high weight for label 1 ===
```

```
  # #
  #  ##
 ##  ###
### ####
```

The answer is (a)

Similar to picture in the a section

File Edit Format Run Options Windows Help

```
# answers.py
# -----
# Licensing Information: You are free to use or extend these projects for
# educational purposes provided that (1) you do not distribute or publish
# solutions, (2) you retain this notice, and (3) you provide clear
# attribution to UC Berkeley, including a link to http://ai.berkeley.edu.
#
# Attribution Information: The Pacman AI projects were developed at UC Berkeley.
# The core projects and autograders were primarily created by John DeNero
# (denero@cs.berkeley.edu) and Dan Klein (klein@cs.berkeley.edu).
# Student side autograding was added by Brad Miller, Nick Hay, and
# Pieter Abbeel (pabbeel@cs.berkeley.edu).
```

```
def q2():
    return 'a'
```

Q3.

```
File Edit Format Run Options Windows Help

# Mira implementation
import util
PRINT = True

class MiraClassifier:
    """
    Mira classifier.

    Note that the variable 'datum' in this code refers to a counter of features
    (not to a raw samples.Datum).
    """
    def __init__(self, legalLabels, max_iterations):
        self.legalLabels = legalLabels
        self.type = "mira"
        self.automaticTuning = False
        self.C = 0.001
        self.legalLabels = legalLabels
        self.max_iterations = max_iterations
        self.initializeWeightsToZero()

    def initializeWeightsToZero(self):
        "Resets the weights of each label to zero vectors"
        self.weights = {}
        for label in self.legalLabels:
            self.weights[label] = util.Counter() # this is the data-structure yo

    def train(self, trainingData, trainingLabels, validationData, validationLabe
        "Outside shell to call your method. Do not modify this method."

        self.features = trainingData[0].keys() # this could be useful for your c

        if (self.automaticTuning):
            Cgrid = [0.002, 0.004, 0.008]
        else:
            Cgrid = [self.C]

        return self.trainAndTune(trainingData, trainingLabels, validationData, v

    def trainAndTune(self, trainingData, trainingLabels, validationData, validat
        """
```

Code with autotune:

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Int
elligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c mira --autotune
Doing classification
-----
data:          digits
classifier:      mira
using enhanced features?:      False
training set size:      100
using automatic tuning for MIRA
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
63 correct out of 100 (63.0%).
Testing...
61 correct out of 100 (61.0%).

```

And code without autotune :

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Int
elligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c mira
Doing classification
-----
data:          digits
classifier:      mira
using enhanced features?:      False
training set size:      100
using default C=0.001 for MIRA
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
64 correct out of 100 (64.0%).
Testing...
58 correct out of 100 (58.0%).

```

Q4.

dataClassifier.py -d digits -c naiveBayes -f -a -t 1000

```
return features
```

```
def enhancedFeatureExtractorDigit(datum):
```

```
    """
```

```
    Your feature extraction playground.
```

```
    You should return a util.Counter() of features  
    for this datum (datum is of type samples.Datum).
```

```
    ## DESCRIBE YOUR ENHANCED FEATURES HERE...
```

```
    ##  
    """
```

```
    has_loop=False
```

```
    run_dfs_hash={}
```

```
    def run_dfs(x,y):
```

```
        if x<0 or x>=DIGIT_DATUM_WIDTH:
```

```
            return False
```

```
        if y<0 or y>=DIGIT_DATUM_HEIGHT:
```

```
            return False
```

```
        if datum.getPixel(x,y)==2 or datum.getPixel(x,y)==1:
```

```
            return True
```

```
        if (x,y) in visited.keys():
```

```
            return True
```

```
        visited[(x,y)]=True
```

```
        return run_dfs(x-1,y) and run_dfs(x+1,y) and run_dfs(x,y-1) and run_dfs(x,y+1)
```

```
    for x in range(DIGIT_DATUM_WIDTH):
```

```
        for y in range(DIGIT_DATUM_HEIGHT):
```

```
            visited={}
```

```
            if datum.getPixel(x,y)==0:
```

```
                if run_dfs(x,y)==True:
```

```
                    has_loop=True
```

```
                    break
```

```
    features = basicFeatureExtractorDigit(datum)
```

```
    features["has_loop"] =has_loop
```

```
    return features
```

```

84 correct out of 100 (84.0%).
PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Int
elligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-d digits -c naiveBayes -f -a -t 1000
Doing classification
-----
data:          digits
classifier:     naiveBayes
using enhanced features?: True
training set size: 1000
using automatic tuning for naivebayes
Extracting features...
Training...
Performance on validation set for k=0.001000: (84.0%)
Performance on validation set for k=0.010000: (83.0%)
Performance on validation set for k=0.050000: (83.0%)
Performance on validation set for k=0.100000: (83.0%)
Performance on validation set for k=0.500000: (82.0%)
Performance on validation set for k=1.000000: (82.0%)
Performance on validation set for k=5.000000: (81.0%)
Performance on validation set for k=10.000000: (81.0%)
Performance on validation set for k=20.000000: (77.0%)
Performance on validation set for k=50.000000: (70.0%)
Validating...
84 correct out of 100 (84.0%).
Testing...
84 correct out of 100 (84.0%).
PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Int

```

Q5.

```

88 correct out of 100 (88.0%).
PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Int
elligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -d pacman
Doing classification
-----
data:          pacman
classifier:     perceptron
using enhanced features?: False
training set size: 100
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
88 correct out of 100 (88.0%).
Testing...
84 correct out of 100 (84.0%).

```

Q6

dataClassifier.py -c perceptron -d pacman -f -g ContestAgent -t 1000 -s 1000

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -d pacman -f -g ContestAgent -t 1000
Doing classification
-----
data:          pacman
classifier:     perceptron
using enhanced features?: True
training set size: 1000
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
95 correct out of 100 (95.0%).
Testing...
94 correct out of 100 (94.0%).

```

dataClassifier.py -c perceptron -d pacman -f -g StopAgent -t 1000 -s 1000

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -d pacman -f -g StopAgent -t 1000
Doing classification
-----
data:          pacman
classifier:     perceptron
using enhanced features?: True
training set size: 1000
Extracting features...
Training...
Starting iteration 0 ...
Starting iteration 1 ...
Starting iteration 2 ...
Validating...
100 correct out of 100 (100.0%).
Testing...
100 correct out of 100 (100.0%).

```

dataClassifier.py -c perceptron -d pacman -f -g FoodAgent -t 1000 -s 1000

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -d pacman -f -g FoodAgent -t 1000
Doing classification
-----
data:          pacman
classifier:      perceptron
using enhanced features?:      True
training set size:      1000
Extracting features...
Training...
Starting iteration  0 ...
Starting iteration  1 ...
Starting iteration  2 ...
Validating...
83 correct out of 100 (83.0%).
Testing...
84 correct out of 100 (84.0%).

```

dataClassifier.py -c perceptron -d pacman -f -g SuicideAgent -t 1000 -s 1000

```

PS C:\Users\Ahmad\Desktop\MS CS\2-Summer-24\CS566-01 202405 - Artificial Intelligence\Final Project\Code\Code> C:\Python27\python.exe dataClassifier.py
-c perceptron -d pacman -f -g SuicideAgent -t 1000
Doing classification
-----
data:          pacman
classifier:      perceptron
using enhanced features?:      True
training set size:      1000
Extracting features...
Training...
Starting iteration  0 ...
Starting iteration  1 ...
Starting iteration  2 ...
Validating...
85 correct out of 100 (85.0%).
Testing...
85 correct out of 100 (85.0%).

```

The final autograder.py result:

Question q1

=====

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

79 correct out of 100 (79.0%).

*** PASS: test_cases\q1\grade.test (4 of 4 points)

*** 79.0 correct (4 of 4 points)

*** Grading scheme:

*** < 70: 0 points

*** >= 70: 4 points

Question q1: 4/4

Question q2

=====

*** PASS: test_cases\q2\grade.test

Question q2: 1/1

Question q3

=====

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

Starting iteration 4 ...

87 correct out of 100 (87.0%).

*** PASS: test_cases\q3\grade.test (6 of 6 points)

*** 87.0 correct (6 of 6 points)

*** Grading scheme:

*** < 80: 0 points

*** >= 80: 6 points

Question q3: 6/6

Question q4

=====

Performance on validation set for k=0.001000: (84.0%)

Performance on validation set for k=0.010000: (83.0%)

Performance on validation set for k=0.050000: (83.0%)

Performance on validation set for k=0.100000: (83.0%)

Performance on validation set for k=0.500000: (82.0%)

Performance on validation set for k=1.000000: (82.0%)

Performance on validation set for k=5.000000: (81.0%)

Performance on validation set for k=10.000000: (81.0%)

Performance on validation set for k=20.000000: (77.0%)

Performance on validation set for k=50.000000: (70.0%)

84 correct out of 100 (84.0%).

*** PASS: test_cases\q4\grade.test (6 of 6 points)

*** 84.0 correct (6 of 6 points)

*** Grading scheme:

*** < 80: 0 points

*** >= 80: 3 points

*** >= 84: 6 points

Question q4: 6/6

Question q5

=====

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

Starting iteration 4 ...

84 correct out of 100 (84.0%).

*** PASS: test_cases\q5\contest.test (2 of 2 points)

*** 84.0 correct (2 of 2 points)

*** Grading scheme:

*** < 70: 0 points

*** >= 70: 2 points

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

Starting iteration 4 ...

72 correct out of 100 (72.0%).

*** PASS: test_cases\q5\suicide.test (2 of 2 points)

*** 72.0 correct (2 of 2 points)

*** Grading scheme:

*** < 70: 0 points

*** >= 70: 2 points

Question q5: 4/4

Question q6

=====

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

94 correct out of 100 (94.0%).

*** PASS: test_cases\q6\contest.test (2 of 2 points)

*** 94.0 correct (2 of 2 points)

*** Grading scheme:

*** < 90: 0 points

*** >= 90: 2 points

Starting iteration 0 ...

Starting iteration 1 ...

Starting iteration 2 ...

Starting iteration 3 ...

85 correct out of 100 (85.0%).

*** PASS: test_cases\q6\suicide.test (2 of 2 points)

*** 85.0 correct (2 of 2 points)

*** Grading scheme:

*** < 80: 0 points

*** >= 80: 2 points

Question q6: 4/4

Finished at 20:04:19

Provisional grades

=====

Question q1: 4/4

Question q2: 1/1

Question q3: 6/6

Question q4: 6/6

Question q5: 4/4

Question q6: 4/4

Total: 25/25